14 Statistical Formula and Statistical Tables

14.1 Statistical Formula

Expected value or mean

$$\mu = E[X] = \begin{cases} \sum_{all\ x} xf(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} xf(x), & \text{if } X \text{ is continuous} \end{cases}$$

Sums of Random Variables

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i]$$

$$\text{Var}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i^2 Var[X_i] + 2\underbrace{\sum\sum}_{i<j} a_i a_j \text{Cov}(X_i, X_j)$$

Variance

$$\text{Var}(X) = E[X^2] - \mu^2 = E[X^2] - \{E[X]\}^2$$

Moments

$$\mu_r' = E[X^r] = \begin{cases} \sum_{R} x^r f(x) & \text{if } X \text{ is discrete} \\ \int_{R} x^r f(x)dx & \text{if } X \text{ is continuous} \end{cases}$$

Covariance

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Moment Generating Function

$$M_X(t) = E[e^{tX}]$$

$$M_X(t) = \begin{cases} \sum_{all\ x} e^{tx} f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{+\infty} e^{tx} f(x)dx & \text{if } X \text{ is continuous} \end{cases}$$

$$\mu_r' = \frac{d^r M_X(t)}{dt^r}\bigg|_{t=0}$$

Maximum likelihood estimator

$$L = L(\theta|x_1, \dots, x_n) = \prod_{i=1}^{n} f(x_i; \theta)$$

or

$$\ln(L) = \sum_{i=1}^{n} \ln\left(f(x_i; \theta)\right)$$

Chi-square

$$X^2 = \sum_{i} \frac{(O_i - E_i)^2}{E_i}$$

The Mann-Whitney/Wilcoxon Rank sum test

Mann and Whitney modified the test and based it on

$$U_1 = W_1 - \frac{n_1(n_1 + 1)}{2}$$

or

$$U_2 = W_2 - \frac{n_2(n_2 + 1)}{2}$$

and

$$U = \min\left(U_1, U_2\right)$$

Mean and Variance

$$\hat{\mu} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad\qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Two Independent Samples (Known variance or large samples):

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has a $N(0,1)$ distribution

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

has approximately $N(0,1)$ distribution

$$Z = \frac{P_1 - P_2 - (\pi_1 - \pi_2)}{\sqrt{\frac{\pi_1(1-\pi_1)}{n_1} + \frac{\pi_2(1-\pi_2)}{n_2}}}$$

has approximately $N(0,1)$ distribution

Small Samples

$$T = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

has a t-distribution with $v = n_1 + n_2 - 2$ df where

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

is a pooled estimate of a common variance σ^2.

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2}$$

has an F distribution with $(n_1 - 1, n_2 - 1)$ df

14.2 Common distributions

14.2.1 Discrete distributions

Name	$f(x)$	$E[X]$	var$[X]$
Geometric if $X \sim Geom(p)$	$f(x) = p(1-p)^{x-1} = pq^{x-1}$	$E[X] = \frac{1}{p}$	$\text{var}(X) = \frac{1-p}{p^2}$
	for $x = 1,2,3,\ldots$ where p is the probability of success on each trial.		
Negative Binomial if $X \sim NegBin(r,p)$	$f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r} = \binom{x-1}{r-1} p^r q^{x-r}$	$E[X] = \frac{r}{p}$	$\text{var}(X) = \frac{r(1-p)}{p^2}$
	for $x = r, r+1, r+2, \ldots$ where r is the number of successes to be achieved and p is the probability of success on each trial.		
Binomial if $X \sim Bin(n,p)$	$f(x) = \binom{n}{x} p^x (1-p)^{n-x} = \binom{n}{x} p^x q^{n-x}$	$E[X] = np$	$\text{var}(X) = np(1-p) = npq$
	for $x = 0,1,2,\ldots,n$ where n is the total number of trials and p is the probability of success on each individual trial.		

Poisson if $X \sim Po(\lambda)$	$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$	$E[X] = \lambda$	$\text{var}(X) = \lambda$
	for $x = 0,1,2,\ldots$ where λ is the average rate of events per interval.		

14.2.2 Continuous distributions

Name	$f(x)$	$E[X]$	var$[X]$
Uniform if $X \sim U(a,b)$	$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{elsewhere} \end{cases}$	$E[X] = \frac{a+b}{2}$	$\text{var}[X] = \frac{(b-a)^2}{12}$
	where a and b represent the lower and upper bounds of the distribution.		
Exponential if $X \sim Exp(\lambda)$	$f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & elsewhere \end{cases}$	$E[X] = \frac{1}{\lambda}$	$\text{var}[X] = \frac{1}{\lambda^2}$
	where λ is the rate parameter, which is the average number of events per unit of time.		
Normal if $X \sim N(\mu, \sigma^2)$	$f(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$	$E[X] = \mu$	$\text{var}[X] = \sigma^2$
	where μ is the mean and σ^2 is the variance parameters.		

14.3 Tables

14.3.1 Standard Normal

Statistical Tables

Standard Normal Probabilities
(The table is based on the area **P** under the standard normal probability curve, below the respective **z**-statistic.)



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-4.0	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00002	0.00002	0.00002	0.00002
-3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
-3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
-3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
-3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
-3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
-3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
-3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
-3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
-3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
-3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
-2.9	0.00187	0.00181	0.00175	0.00169	0.00164	0.00159	0.00154	0.00149	0.00144	0.00139
-2.8	0.00256	0.00248	0.00240	0.00233	0.00226	0.00219	0.00212	0.00205	0.00199	0.00193
-2.7	0.00347	0.00336	0.00326	0.00317	0.00307	0.00298	0.00289	0.00280	0.00272	0.00264
-2.6	0.00466	0.00453	0.00440	0.00427	0.00415	0.00402	0.00391	0.00379	0.00368	0.00357
-2.5	0.00621	0.00604	0.00587	0.00570	0.00554	0.00539	0.00523	0.00508	0.00494	0.00480
-2.4	0.00820	0.00798	0.00776	0.00755	0.00734	0.00714	0.00695	0.00676	0.00657	0.00639
-2.3	0.01072	0.01044	0.01017	0.00990	0.00964	0.00939	0.00914	0.00889	0.00866	0.00842
-2.2	0.01390	0.01355	0.01321	0.01287	0.01255	0.01222	0.01191	0.01160	0.01130	0.01101
-2.1	0.01786	0.01743	0.01700	0.01659	0.01618	0.01578	0.01539	0.01500	0.01463	0.01426
-2.0	0.02275	0.02222	0.02169	0.02118	0.02067	0.02018	0.01970	0.01923	0.01876	0.01831
-1.9	0.02872	0.02807	0.02743	0.02680	0.02619	0.02559	0.02500	0.02442	0.02385	0.02330
-1.8	0.03593	0.03515	0.03438	0.03362	0.03288	0.03216	0.03144	0.03074	0.03005	0.02938
-1.7	0.04456	0.04363	0.04272	0.04181	0.04093	0.04006	0.03920	0.03836	0.03754	0.03673
-1.6	0.05480	0.05370	0.05262	0.05155	0.05050	0.04947	0.04846	0.04746	0.04648	0.04551
-1.5	0.06681	0.06552	0.06425	0.06301	0.06178	0.06057	0.05938	0.05821	0.05705	0.05592
-1.4	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07214	0.07078	0.06944	0.06811
-1.3	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08691	0.08534	0.08379	0.08226
-1.2	0.11507	0.11314	0.11123	0.10935	0.10749	0.10565	0.10383	0.10204	0.10027	0.09852
-1.1	0.13566	0.13350	0.13136	0.12924	0.12714	0.12507	0.12302	0.12100	0.11900	0.11702
-1.0	0.15865	0.15625	0.15386	0.15150	0.14917	0.14686	0.14457	0.14231	0.14007	0.13786
-0.9	0.18406	0.18141	0.17878	0.17618	0.17361	0.17105	0.16853	0.16602	0.16354	0.16109

-0.8	0.21185	0.20897	0.20611	0.20327	0.20045	0.19766	0.19489	0.19215	0.18943	0.18673
-0.7	0.24196	0.23885	0.23576	0.23269	0.22965	0.22663	0.22363	0.22065	0.21769	0.21476
-0.6	0.27425	0.27093	0.26763	0.26434	0.26108	0.25784	0.25462	0.25143	0.24825	0.24509
-0.5	0.30853	0.30502	0.30153	0.29805	0.29460	0.29116	0.28774	0.28434	0.28095	0.27759
-0.4	0.34457	0.34090	0.33724	0.33359	0.32997	0.32635	0.32276	0.31917	0.31561	0.31206
-0.3	0.38209	0.37828	0.37448	0.37070	0.36692	0.36317	0.35942	0.35569	0.35197	0.34826
-0.2	0.42074	0.41683	0.41293	0.40904	0.40516	0.40129	0.39743	0.39358	0.38974	0.38590
-0.1	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43250	0.42857	0.42465
-0.0	0.50000	0.49601	0.49202	0.48803	0.48404	0.48006	0.47607	0.47209	0.46811	0.46414

14.3.2 Binomial

Binomial probabilities $b(x;n,p)=\binom{n}{x}p^x q^{n-x} \quad x=0,1,2,...,n$

		p										
	x	0.01	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
n=14	0	0.8687	0.4877	0.2288	0.1028	0.0440	0.0178	0.0068	0.0024	0.0008	0.0002	0.0001
	1	0.1229	0.3593	0.3559	0.2539	0.1539	0.0832	0.0407	0.0181	0.0073	0.0027	0.0009
	2	0.0081	0.1229	0.2570	0.2912	0.2501	0.1802	0.1134	0.0634	0.0317	0.0141	0.0056
	3	0.0003	0.0259	0.1142	0.2056	0.2501	0.2402	0.1943	0.1366	0.0845	0.0462	0.0222
	4		0.0037	0.0349	0.0998	0.1720	0.2202	0.2290	0.2022	0.1549	0.1040	0.0611
	5		0.0004	0.0078	0.0352	0.0860	0.1468	0.1963	0.2178	0.2066	0.1701	0.1222
	6			0.0013	0.0093	0.0322	0.0734	0.1262	0.1759	0.2066	0.2088	0.1833
	7			0.0002	0.0019	0.0092	0.0280	0.0618	0.1082	0.1574	0.1952	0.2095
	8				0.0003	0.0020	0.0082	0.0232	0.0510	0.0918	0.1398	0.1833
	9					0.0003	0.0018	0.0066	0.0183	0.0408	0.0762	0.1222
	10						0.0003	0.0014	0.0049	0.0136	0.0312	0.0611
	11							0.0002	0.0010	0.0033	0.0093	0.0222
	12								0.0001	0.0005	0.0019	0.0056
	13									0.0001	0.0002	0.0009
	14											0.0001
n=15	0	0.8601	0.4633	0.2059	0.0874	0.0352	0.0134	0.0047	0.0016	0.0005	0.0001	
	1	0.1303	0.3658	0.3432	0.2312	0.1319	0.0668	0.0305	0.0126	0.0047	0.0016	0.0005
	2	0.0092	0.1348	0.2669	0.2856	0.2309	0.1559	0.0916	0.0476	0.0219	0.0090	0.0032
	3	0.0004	0.0307	0.1285	0.2184	0.2501	0.2252	0.1700	0.1110	0.0634	0.0318	0.0139
	4		0.0049	0.0428	0.1156	0.1876	0.2252	0.2186	0.1792	0.1268	0.0780	0.0417
	5		0.0006	0.0105	0.0449	0.1032	0.1651	0.2061	0.2123	0.1859	0.1404	0.0916
	6			0.0019	0.0132	0.0430	0.0917	0.1472	0.1906	0.2066	0.1914	0.1527
	7			0.0003	0.0030	0.0138	0.0393	0.0811	0.1319	0.1771	0.2013	0.1964
	8				0.0005	0.0035	0.0131	0.0348	0.0710	0.1181	0.1647	0.1964
	9				0.0001	0.0007	0.0034	0.0116	0.0298	0.0612	0.1048	0.1527
	10					0.0001	0.0007	0.0030	0.0096	0.0245	0.0515	0.0916
	11						0.0001	0.0006	0.0024	0.0074	0.0191	0.0417
	12							0.0001	0.0004	0.0016	0.0052	0.0139
	13								0.0001	0.0003	0.0010	0.0032
	14										0.0001	0.0005
	15											

n=19	0	0.8262	0.3774	0.1351	0.0456	0.0144	0.0042	0.0011	0.0003	0.0001		
	1	0.1586	0.3774	0.2852	0.1529	0.0685	0.0268	0.0093	0.0029	0.0008	0.0002	
	2	0.0144	0.1787	0.2852	0.2428	0.1540	0.0803	0.0358	0.0138	0.0046	0.0013	0.0003
	3	0.0008	0.0533	0.1796	0.2428	0.2182	0.1517	0.0869	0.0422	0.0175	0.0062	0.0018
	4		0.0112	0.0798	0.1714	0.2182	0.2023	0.1491	0.0909	0.0467	0.0203	0.0074
	5		0.0018	0.0266	0.0907	0.1636	0.2023	0.1916	0.1468	0.0933	0.0497	0.0222
	6		0.0002	0.0069	0.0374	0.0955	0.1574	0.1916	0.1844	0.1451	0.0949	0.0518
	7			0.0014	0.0122	0.0443	0.0974	0.1525	0.1844	0.1797	0.1443	0.0961
	8			0.0002	0.0032	0.0166	0.0487	0.0981	0.1489	0.1797	0.1771	0.1442
	9				0.0007	0.0051	0.0198	0.0514	0.0980	0.1464	0.1771	0.1762
	10				0.0001	0.0013	0.0066	0.0220	0.0528	0.0976	0.1449	0.1762
	11					0.0003	0.0018	0.0077	0.0233	0.0532	0.0970	0.1442
	12						0.0004	0.0022	0.0083	0.0237	0.0529	0.0961
	13						0.0001	0.0005	0.0024	0.0085	0.0233	0.0518
	14							0.0001	0.0006	0.0024	0.0082	0.0222
	15								0.0001	0.0005	0.0022	0.0074
	16									0.0001	0.0005	0.0018
	17										0.0001	0.0003
	18											
	19											

14.3.3 Wilcoxon Signed Ranks Test

n	Two-Tailed Test		One-Tailed Test	
	$\alpha = .05$	$\alpha = .01$	$\alpha = .05$	$\alpha = .01$
5	--	--	0	--
6	0	--	2	--
7	2	--	3	0
8	3	0	5	1
9	5	1	8	3
10	8	3	10	5
11	10	5	13	7
12	13	7	17	9
13	17	9	21	12
14	21	12	25	15
15	25	15	30	19
16	29	19	35	23
17	34	23	41	27
18	40	27	47	32
19	46	32	53	37
20	52	37	60	43
21	58	42	67	49
22	65	48	75	55
23	73	54	83	62
24	81	61	91	69
25	89	68	100	76
26	98	75	110	84
27	107	83	119	92
28	116	91	130	101
29	126	100	140	110
30	137	109	151	120

14.3.4 Mann-Whitney/Wilcoxon rank-sum two sample test

5% critical region: $U \leqslant$ tabulated value

n_L \ n_S	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
2	-	-	-	-	-	-	0	0	0	0	1	1	1	1	1	2	2	2	2	3	3	3	3	3
3	-	-	-	0	1	1	2	2	3	3	4	4	5	5	6	6	7	8	8	9	9	10	10	10
4	-	-	-	1	2	3	4	4	5	6	7	8	9	10	11	11	12	13	14	15	16	17	17	18
5	-	-	-	-	3	5	6	7	8	9	11	12	13	14	15	17	18	19	20	22	23	24	25	27
6	-	-	-	0	1	3	6	8	10	11	13	14	16	17	19	21	22	24	25	27	29	30	32	33	35
7	-	-	-	0	1	3	6	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44
8	-	-	-	1	2	4	6	15	15	17	19	22	24	26	29	31	34	36	38	41	43	45	48	50	53
9	-	-	0	1	3	5	7	9	20	20	23	26	28	31	34	37	39	42	45	48	50	53	56	59	62
10	-	-	0	2	4	6	9	11	13	26	26	29	33	36	39	42	45	48	52	55	58	61	64	67	71
11	-	-	0	2	5	7	10	13	16	18	29	33	37	40	44	47	51	55	58	62	65	69	73	76	80
12	-	-	1	3	6	9	12	15	18	21	24	33	41	45	49	53	57	61	65	69	73	77	81	85	89
13	-	-	1	3	7	10	13	17	20	24	27	31	38	50	54	59	63	67	72	76	80	85	89	94	98
14	-	-	1	4	7	11	15	18	22	26	30	34	42	46	59	64	69	74	78	83	88	93	98	102	107
15	-	-	2	5	8	12	16	20	24	29	33	37	45	50	55	70	75	80	85	90	96	101	106	111	117
16	-	-	2	5	9	13	18	22	27	31	36	41	49	54	60	65	81	86	92	98	103	109	115	120	126
17	-	-	2	6	10	15	19	24	29	34	39	44	49	54	60	65	93	99	105	111	117	123	129	135	141
18	-	-	2	6	11	16	21	26	31	37	42	47	53	58	64	70	75	106	112	119	125	132	138	145	152
19	0	3	7	12	17	22	28	33	39	45	51	57	63	69	74	81	87	99	119	126	133	140	147	154	161
20	0	3	8	13	18	24	30	36	42	48	54	60	67	73	79	86	92	99	134	141	149	156	163	170	178
21	0	3	8	14	19	25	32	38	44	51	58	64	71	78	84	91	98	105	112	150	157	165	173	180	188
22	0	4	9	14	21	27	34	40	47	54	61	68	75	82	89	96	104	111	118	125	166	174	182	189	197
23	0	4	9	15	22	29	35	43	50	57	64	72	79	87	94	102	109	117	125	132	140	183	192	200	208
24	0	4	10	16	23	30	37	45	52	60	68	75	83	91	99	107	115	123	131	139	147	155	201	209	217
25	0	5	10	17	24	32	39	47	55	63	71	79	87	96	104	112	121	129	138	146	155	163	172	217	226

1%

equal sample sizes

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
5%	-	-	-	-	0	2	5	8	13	17	23	30	37	45	55	64	75	87	99	113	127	142	158	175	192	211
1%	-	-	-	-	-	0	2	4	7	11	16	21	27	34	42	51	60	70	81	93	105	118	133	148	164	180

n	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
5%	230	250	272	294	317	341	365	391	418	445	473	503	533	564	596	628	662	697	732	769	806	845	884	924	965
1%	198	216	235	255	276	298	321	344	369	394	420	447	475	504	533	564	595	627	660	694	729	765	802	839	877

12.3.5 The Chi-square distribution

The Chi-square distribution

Rejection zone

df	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	---	---	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401

df	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40. 289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

12.3.6 Tukey's Wholly Significant Difference

Table 11

TUKEY'S WHOLLY SIGNIFICANT DIFFERENCE (5% LEVEL)

The table gives the values of the Wholly Significant Difference/s ($\bar{x}$) where:

W.S.D. = 'Wholly Significant Difference' between means at 5% level.
s = Estimate of standard deviation
s ($\bar{x}$) = Estimate of standard error of the means = $s/\sqrt{n}$
k = Number of means under test
ν = Degrees of freedom associated with s^2, the estimate of error variance
n = Number of readings in each mean

ν \ k	2	3	4	5	6	7	8	9	10	12	15	20	30	60
1	21.96	28.80	34.56	39.60	44.28	48.78	53.10	55.62	57.96	62.46	69.12	77.58	91.98	116.64
2	6.83	8.54	9.88	10.98	11.95	12.87	13.73	14.27	14.76	15.68	17.02	18.67	21.41	26.11
3	4.89	6.00	6.84	7.50	8.07	8.60	9.08	9.41	9.69	10.22	10.98	11.89	13.40	15.96
4	4.19	5.10	5.76	6.27	6.70	7.10	7.45	7.70	7.92	8.31	8.88	9.55	10.62	12.47
5	3.86	4.66	5.24	5.68	6.04	6.37	6.66	6.88	7.06	7.39	7.86	8.41	9.28	10.77
6	3.64	4.38	4.91	5.31	5.63	5.92	6.17	6.37	6.53	6.82	7.23	7.70	8.45	9.73
7	3.50	4.20	4.70	5.06	5.35	5.62	5.85	6.03	6.17	6.44	6.81	7.24	7.91	9.05
8	3.41	4.08	4.55	4.89	5.17	5.41	5.62	5.79	6.00	6.18	6.53	6.92	7.53	8.57
9	3.34	3.98	4.44	4.76	5.03	5.26	5.46	5.62	5.75	5.98	6.31	6.68	7.25	7.99
10	3.28	3.91	4.35	4.66	4.91	5.13	5.32	5.48	5.61	5.83	6.14	6.49	7.02	7.94
12	3.19	3.80	4.22	4.52	4.75	4.96	5.13	5.28	5.40	5.61	5.90	6.22	6.70	7.54
15	3.11	3.69	4.09	4.37	4.60	4.79	4.95	5.09	5.20	5.39	5.66	5.95	6.39	7.14
20	3.04	3.60	3.98	4.25	4.45	4.63	4.78	4.91	5.02	5.19	5.44	5.71	6.11	6.79
30	2.97	3.51	3.87	4.12	4.32	4.48	4.61	4.74	4.84	5.00	5.23	5.47	5.83	6.44
60	2.90	3.41	3.76	4.00	4.18	4.33	4.45	4.57	4.66	4.81	5.02	5.24	5.56	6.09
∞	2.82	3.32	3.66	3.88	4.04	4.18	4.29	4.40	4.49	4.63	4.82	5.01	5.29	5.76

See page 53 for an example on the use of this table.

Applied Statistics for Engineers, page 179. W. Volk (McGraw-Hill)
(Modification of tables in ***The Problem of Multiple Comparison***, mimeographed publication 1953, J. W. Tukey)